

Abhay Pratap Singh

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Professional Summary

CS undergrad specialising in frontend engineering and autonomous systems, with active coursework in AI/LLM engineering covering Python, Docker, LLM fundamentals, and agentic AI systems. Proficient in English (written and verbal). National Rank 8 at Hack2Future (IIIT Dharwad); winner of Technotsav'25 by INAE; James Dyson Award entrant. Microsoft Learn Student Ambassador and open-source contributor via GirlScript Summer of Code. Proficient in React, TypeScript, REST APIs, LangGraph, ROS2, and PyTorch.

Technical Skills

Frontend: React.js, JavaScript (ES6+), TypeScript, HTML5, CSS3, Bootstrap, Backpack React Library, Responsive Design
Programming: Java, C, Python, LangGraph **Tools:** Git, RESTful APIs, Docker, Postman, Hugging Face, ROS2, PyTorch

Education

Cambridge Institute of Technology, BE in Computer Science — GPA: 3.5/4.0 Sept 2024 – June 2028

- **Coursework:** Data Structures, Web Development, Software Engineering, Database Systems

Experience

Frontend Developer Intern, Zetheta Algorithms Private Limited April 2026 – Present

- Building production-grade React/TypeScript frontend applications across four live projects: Real Time Stock Screener, Multi Step Loan Application Form, Portfolio Analytics Dashboard, and Interactive Financial Education Module Builder.
- Developing responsive, dynamic UIs with REST API integration, real-time data updates, and modular component architecture.

National Rank 8 — Hack2Future Hackathon, IIIT Dharwad April 2026

- Built a DRL-based autonomous robot navigation system (TD3, ROS2, Gazebo, LiDAR, odometry); owned frontend visualisation dashboard and backend training pipeline.
- Achieved 100% training success rate across randomised obstacle environments; ranked 8th nationally.

Technical Team Member, Microsoft Learn Student Ambassador — CIT Bangalore Oct 2025 – Present

- Organised technical workshops on Microsoft and AI tools; co-built an AI Poster Generator automating design creation with AI models.

Open Source Contributor, GirlScript Summer of Code Aug – Oct 2024

- Contributed to 5+ open-source projects (React, JavaScript); collaborated with 50+ developers using Git and professional code review workflows.

Certificates

- **NPTEL Java** (July–Aug 2025) — Secured 81% in examination.
- **Oracle Generative AI** (Oracle University) — LLMs, token mechanics, and agentic AI workflows.
- **Postman API Fundamentals** — API design, testing, status codes, and HTTP methods (GSSOC 2025).

Projects

- **Real Time Stock Screener** — React + REST APIs app to fetch, filter, and display live stock data with dynamic UI updates.
- **Autonomous Mobile Robot Navigation** — DRL (TD3) navigation with LiDAR and odometry in ROS2/Gazebo using PyTorch.
- **N8N Automation** — Automated pipeline to discover hackathons/ideathons in Bangalore and send email alerts.
- **AI Poster Generator** — AI-powered design automation tool built with the MLSA technical team.

Achievements

- **Amazon Machine Learning Summer School 2026** — Cleared Round 1 (2026)
- **Hack2Future National Rank 8** — IIIT Dharwad, autonomous robot navigation using DRL (April 2026)
- **James Dyson Award** — Entry approved and published in international online gallery (Jul 2025)
- **Technotsav'25 Winner** — 1st Place, Indian National Academy of Engineering (Jan 2025)
- **Zenithx Winner** — 1st Place, Zenithx Hackathon
- **Google Big Code Challenge 2026** — Cleared Qualifier Round; advanced to Round 1 among top 15,000 students nationally (Mar 2026)